香港中文大學 The Chinese University of Hong Kong | 文學院 Faculty of Arts

大學教育資助委員會 University Grants Committee

Workshop 4

Sentiment Analysis in Python

Session Lead

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Goals of Today's Workshop

Run pretrained classifier

Execute emotion analysis model in Google Colab environment

Generate results file

Create output with emotion labels and confidence scores

Visualise emotion distribution

Compare emotional patterns between two video comment datasets

- **What We're NOT Doing**

  No model training today; we're leveraging pretrained capabilities

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Dataset Overview

YouTube comments from **Islam, Judaism, and Christianity – A Conversation**

https://www.youtube.com/watch?v=INIG636SnU4

Ethics

- Educational use only
- Text-only analysis (no user profiling)
- Do not share raw dataset outside workshop

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Session Roadmap

1. Core concepts & reading outputs
2. Hands-on coding in Colab (Steps 1–4)
3. Wrap-up, limitations & Q&A

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Sentiment Analysis

Concepts and theoretical framing

Hussein Hassan

Third-year PhD Candidate

Department of Cultural and Religious Studies, CUHK

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Today's roadmap

1. Quick warm-up
2. A short audience activity
3. What sentiment analysis is
4. Where it is used
5. Why it matters for Humanities
6. Hands-on

   Hugging Face +
   Google Colab

How are you feeling today?

Describe your current mood in 1–3 words on *Mentimeter*

(*anonymous, no personal details*)

https://www.menti.com/al6977o4ev2y

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Activity: Let's do sentiment analysis

We will use 3 short responses from the audience.

For each text, we will apply three tasks:

1. Simplest task — **Judge if the sentiment of this text is positive or negative**
2. More complex task — **Rank the sentiment of this text from 1 to 5**
3. Advanced task — **Label the sentiment of this text (Anger, Fear, Joy, Love, Sadness, Surprise, and Neutral)**

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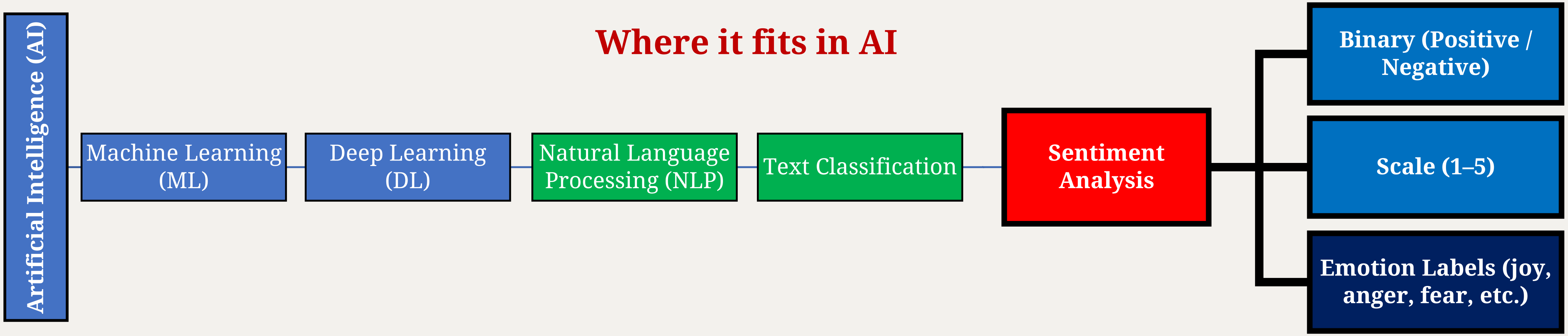
What we just did: Sentiment Analysis

Sentiment analysis is an NLP task that classifies emotional tone in text.



Three common task types:

1. Positive / Negative (binary)
2. 1-5 rating (scale)
3. **Emotion labels (joy, anger, fear, etc.)**

Model output:

label + score (confidence, not intensity)

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Where is sentiment analysis used?

Social media & public reaction tracking

Reviews and customer feedback

Media/marketing:

compare audience reception across campaigns or over time

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Why it matters for Humanities

1. **Focus:** meaning, context, genre
2. **Use cases:** reception/discourse analysis, narrative mood, cultural trends
3. **Key cautions:** sarcasm, negation, mixed emotions, missing context

Use the results as clues, not final answers

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Hands-on Sentiment Analysis

Hugging Face + Google Colab

Mohamed Byomi

First-year PhD Student

Department of Cultural and Religious Studies, CUHK

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Hugging Face

AI community:

“The platform where the machine learning community collaborates on models, datasets, and applications”

- **Sign up:**

  https://huggingface.co/

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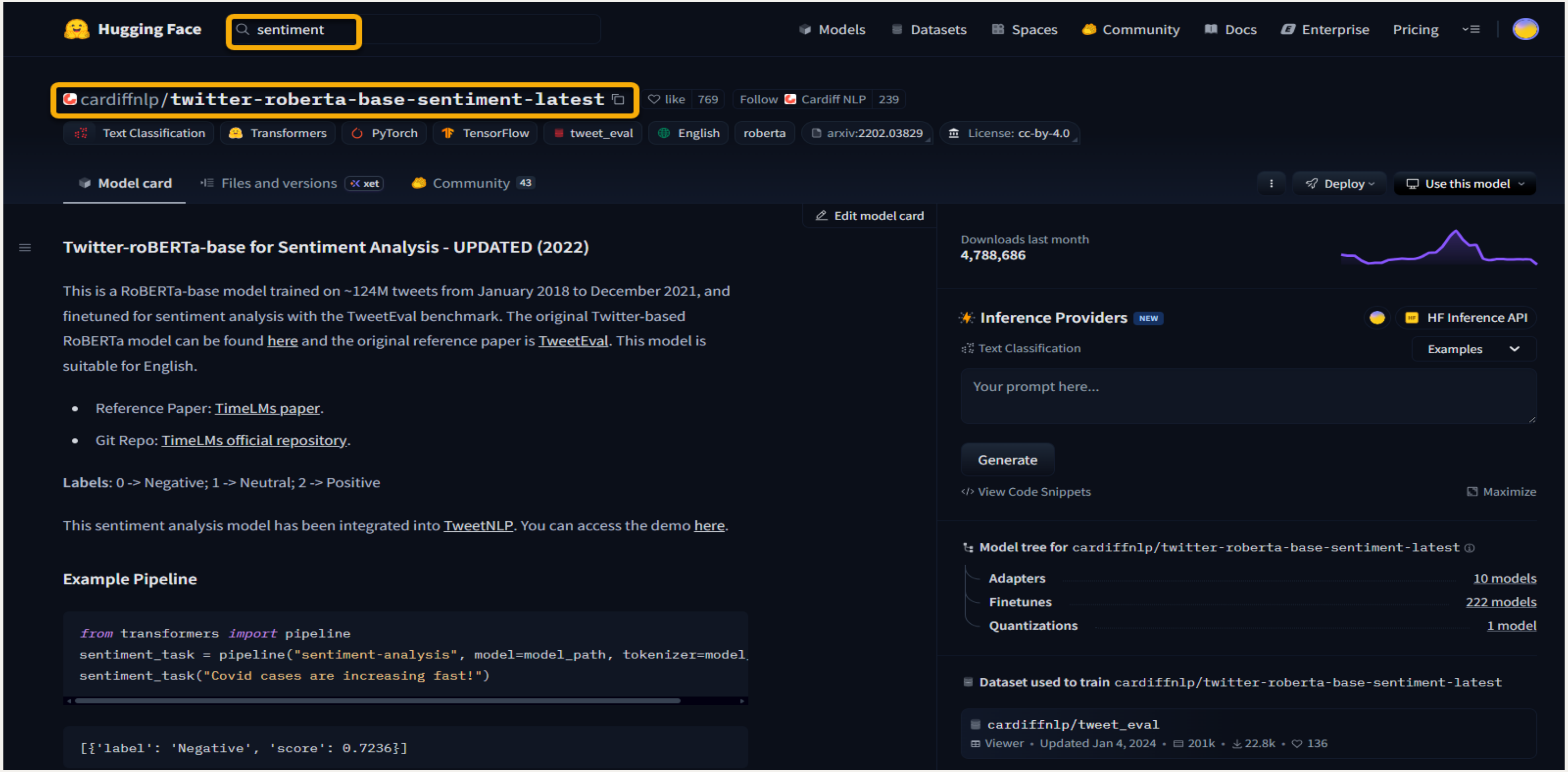
Hugging Face Models



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Hands-On Setup (Step 0)

Open notebook

Launch provided Google Colab notebook

http://tiny.cc/7660101

Verify data

Confirm downloaded CSV file

http://tiny.cc/cuaisent

Preview content

Check first 3–5 rows contain comment text and video identifiers

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Load Libraries & Model (Step 1)

Step 1

```python
# @title Step 1

# Load modules
from transformers import pipeline
from matplotlib import pyplot as plt
import seaborn as sns
import csv
import pandas as pd

# Load the fine-tuned BERT Mini sentiment analysis model
sentiment_analysis = pipeline(
    "text-classification",
    model="j-hartmann/emotion-english-distilroberta-base",
    truncation=True,
    max_length=512
)
```

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Run Analysis (Step 2)

Step 2

```python
# @title Step 2

result = sentiment_analysis("I feel happy!")
print(result)
```

```
[{'label': 'joy', 'score': 0.9896815419197083}]
```

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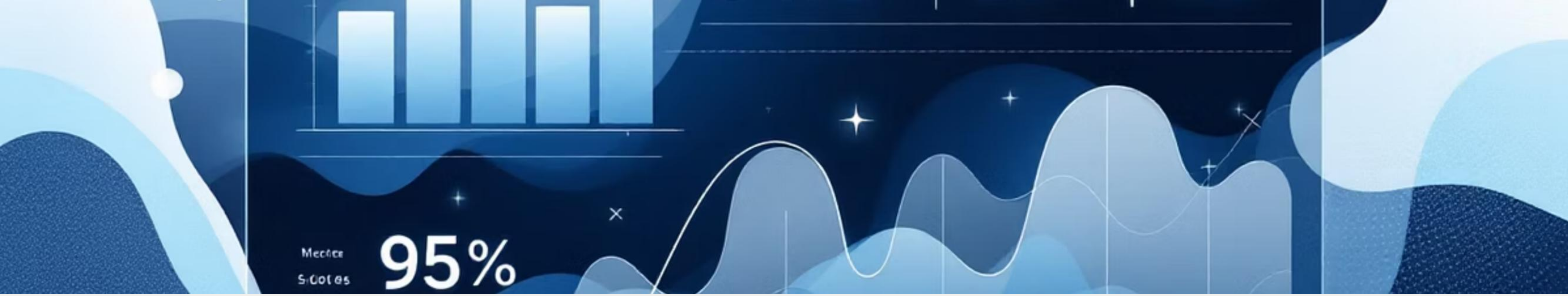

Interpreting Model Output

Label

Predicted emotion category (e.g., joy, anger, fear)

Score

Model's confidence or probability value

> **Important:** Score is not emotional intensity, not psychological diagnosis, not objective truth

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Read data, run analysis, save output (Step 3)

Step 3

```python
# @title Step 3

# Analyze sentiment for each row in your CSV file

results = []

with open('Day_1_workshop_3_sentiment_analysis_in_python.csv', mode='r', newline='', encoding='utf-8-sig') as file:
    reader = csv.reader(file)
    next(reader)  # skip the header

    # Iterate through each row in the CSV file
    for row in reader:

        # Apply sentiment analysis
        sentiment_result = sentiment_analysis(row[1])

        # Extract sentiment label
        sentiment_label = sentiment_result[0]['label']

        # Append the result as a dictionary
        results.append({'text': row[1], 'sentiment_label': sentiment_label, 'video': row[-1]})

        # Print results
        print('text:', row[1], 'sentiment_label:', sentiment_label)


# Create a DataFrame from the results
df = pd.DataFrame(results)

# Save the DataFrame to a single CSV file
df.to_csv("sentiment_results.csv", index=False, encoding="utf-8-sig")
```

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Visualise Results (Step 4)

Step 4

```
# @title Step 4

# Define the desired order of sentiment labels
sentiment_order = ['joy', 'love', 'surprise', 'fear', 'anger', 'sadness', 'neutral']

# Ensure the sentiment_label is a categorical type with the specified order
df['sentiment_label'] = pd.Categorical(df['sentiment_label'], categories=sentiment_order, ordered=True)

# Get unique videos
unique_videos = df['video'].unique()

# Set some styles
sns.set_style("whitegrid")
sns.set_context("talk")

# Create subplots
fig, axs = plt.subplots(len(unique_videos), 1, figsize=(8, 6 * len(unique_videos)), sharey=True)

# Loop through each unique video
for i, video in enumerate(unique_videos):
    df_video = df[df['video'] == video]
    sentiment_counts = df_video['sentiment_label'].value_counts().reindex(sentiment_order, fill_value=0)

    sns.barplot(x=sentiment_counts.values, y=sentiment_counts.index, ax=axs[i])

    axs[i].set_title(f'Distribution of Sentiment Labels for Video: {video}', fontsize=14, fontweight='bold')
    axs[i].set_xlabel('Count', fontsize=12)
    axs[i].set_ylabel('Sentiment Label', fontsize=12)
    axs[i].grid(axis='x', linestyle='--', alpha=0.7)

    for container in axs[i].containers:
        axs[i].bar_label(container, fmt='%d')

# Adjust layout
plt.tight_layout()
plt.show()
```

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Common Limitations

Be careful of overinterpretation due to:

Sarcasm / Irony

Model struggles with tone reversal

Negation patterns

E.g., “not good” confuses simple classifiers

Mixed emotions

Single comment expresses multiple feelings

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Recap & Q&A

Workflow Recap

1. **Load model**
2. **Test**
3. **Run on dataset**
4. **Visualise**

Key Discussion

Do these labels reflect actual emotion or just stylistic tone?

Q&A

Questions and discussion

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